

A. Método 1: Bisseccao

$$[1, 2] \quad f(x) = -x^3 - 2x + 10$$

$$f(1) = -1 - 2 + 10 = 7$$

$$f(2) = -8 - 4 + 10 = -2$$

$$x_1 = \frac{1+2}{2} = 1.5 \quad f(1.5) = -1.5^3 - 2 \cdot 1.5 + 10 = 3.625$$

$$[1.5, 2]$$

$$x_2 = \frac{1.5+2}{2} = 1.75 \quad f(1.75) = -1.75^3 - 2 \cdot 1.75 + 10$$

$$f(1.75) = 1.340625$$

$$1^{\circ} \text{ iteracao: } 3.625$$

$$2^{\circ} \text{ iteracao: } 1.340625$$

B. Método 2: Newton - Raphson

$$f(i) = -i^3 - 2i + 10$$

$$f'(i) = -3i^2 - 2$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$i_0 = 1,5$$

$$f(1,5) = -1,5^3 - 2 \cdot 1,5 + 10 = 3,625$$

$$f'(1,5) = -3 \cdot 1,5^2 - 2 = -8,75$$

$$x_1 = 1,5 - \frac{3,625}{-8,75} = 1,914286$$

$$f(1,914286) = -1,914286^3 - 2 \cdot 1,914286 + 10$$

$$f(1,914286) = -0,843455$$

$$f'(1,914286) = -3 \cdot 1,914286^2 - 2 = -12,993472$$

$$x_2 = 1,914286 - \frac{-0,843455}{-12,993472} = 1,849372$$

$$1^{\text{a}} \text{ iteración: } 1,914286$$

$$2^{\text{a}} \text{ iteración: } 1,849372$$

C. Método 3: Secantes

$$f(x) = -x^3 - 2x + 10 \quad x_0 = 1 \quad x_1 = 2$$

$$f(1) = -1 - 2 + 10 = 7$$

$$f(2) = -8 - 4 + 10 = -2$$

$$x_{k+1} = x_k - \frac{f(x_k) - x_k f(x_{k-1})}{f(x_k) - f(x_{k-1})}$$

$$x_2 = \frac{1 \cdot (-2) - 2 \cdot 7}{-2 - 7} = \frac{-16}{-9} = 1.777778$$

$$f(1.777778) = -1.777778^3 - 2 \cdot 1.777778 + 10$$

$$f(1.777778) = 0.325786$$

$$x_3 = \frac{2 \cdot 0.325786 - 1.777778 \cdot (-2)}{0.325786 - (-2)} = 1.842718$$

$$x_3 = \frac{5.207128}{2.825786} = 1.842718$$

$$1^{\text{a}} \text{ iteración: } 1.777778$$

$$2^{\text{a}} \text{ iteración: } 1.842718$$